LIRnet: Lightweight Hyperspectral Image Classification Based on Information Redistribution

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***Abstract*— Deep learning has received much attention in hyperspectral image (HSI) classification. However, most deep learning methods design relatively complex feature extraction and processing network modules for the characteristics of HSIs, which may not be necessary for relatively simple patch-based HSI classification tasks. The complex network structure and high feature channel dimension lead to large computational complexities, which limit the practical applicability of HSI. In this article, an elegant lightweight HSI classification-based information redistribution network (LIRnet) is proposed to separate and reaggregate the feature information to achieve feature information homogenization and extract discriminative feature information, respectively. The classification performance of LIRnet is better than that of existing methods on three different datasets in different scenarios, which proves its effectiveness. In addition, experiments on embedded devices verify the computational efficacy of LIRnet.**

***Index Terms*— Hyperspectral image (HSI) classification, information redistribution model, lightweight model, weighted sparse attention module (WSAM).**

I. Introduction

HYPERSPECTRAL images, comprising dozens or even hundreds of contiguous bands capturing reflectance spectrographic information from the objects, offer a wealth of discriminative data crucial for land use and land objects' recognition [1], [2], [3]. However, due to the complexity of high-dimensional data and the insufficient number of labeled samples in hyperspectral image (HSI), there are great challenges in practical applications based on HSI [4].

In recent years, a large number of classification algorithms for HSI have been proposed, with deep learning methods attracting wide attention due to their powerful ability to extract fine features. These methods can be broadly categorized into basic convolutional neural network (CNN) models, spectral–spatial CNN models, multifeature fusion CNN models, and advanced CNN architectures. For instance, the CNN with deep pixel-pair features leveraged pixel-pair inputs to learn detailed features, providing a foundational approach for HSI classification [5]. Various models have been developed to capture both spectral and spatial information. Zhang et al. [6] proposed the interleaving perception CNN, which integrated heterogeneous information for enhanced classification performance. In [7], the original data underwent randomized principal component analysis, and then a CNN was used to extract finer details. Mei et al. [8] devised spectral–spatial attention networks with a CNN architecture to gain comprehensive spatial and spectral information. Lee and Kwon [9] used the contextual deep CNN to effectively extract spectral–spatial information, improving the model's ability to interpret HSI data. Zhang et al. [10] proposed the diverse region CNN to process the target patch and the surrounding patches to capture interactive contextual information. To alleviate the problem of limited data samples, Zhang et al. [11] proposed an HSI classification method based on dense pyramidal convolution and multifeature fusion. This method enhanced feature extraction by combining multiple sources of information. Another approach by Tu et al. [12] involved a method for classifying HSI based on residual dense and dilated convolution, which strengthened the combination of shallow and deep features from the network. In efforts to fully exploit local spectral structural information, Zhang et al. [13] developed a 3-D CNN framework for tree species categorization. Hu et al. [14] introduced the hybrid CNN combining multiscale 3-D depthwise separable convolution to address issues of insufficient spectral and spatial information extraction. Ye et al. [15] designed a multiscale convolutional network to exploit discriminative information from spectral data, improving classification accuracy. Zhao et al. [16] built grouped separable convolution, incorporating Gabor filters to control the kernels and use orientation and scale information for feature extraction. Gao et al. [17] proposed an adversarial complementary learning strategy aimed at extracting superior spectral information from HSI and improving classification representation. They also developed an improved transformer baseline for HSI classification, using cross-scale mixing attention to balance global and local dependencies [18]. A spatial–spectral perception network was used by Gao et al. [4] to explore the

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spatial distribution relationship and spectral correlation of land objects, achieving efficient feature expression.

Although CNN-based HSI classification models have achieved satisfactory results, they often struggle with capturing long-range dependencies. To address this limitation, attention-based transformer models have gained popularity due to their ability to effectively extract long-range relationships. These models, based on self-attention mechanisms [19], [20], [21], have shown significant success across various machine learning applications. In the realm of HSI classification, these transformer-based methods can be categorized into several types: Liu et al. [22] introduced the central attention network, which integrated spatial and spectral information to improve classification performance. Building on spatial and spectral information extraction, He et al. [23] proposed the bidirectional encoder, which processed a pixel sequence comprising the target pixel and its neighboring pixels by embedding positional information, further enhancing the model's ability to use contextual information. Pu et al. [24] combined attention mechanisms with CNNs to construct the multilevel feature network, leveraging the strengths of both the approaches to capture detailed features and contextual information. Expanding on the hybrid model concept, Zhang et al. [25] developed a graph information aggregation framework for cross-domain few-shot learning, which used graph information to enhance classification performance and handle limited data scenarios. To address the challenge of hyperparameter optimization, Xue et al. [26] proposed an attention-based second-order pooling network, which simplified the hyperparameter tuning process while maintaining high classification accuracy. Further refining this approach, Xue et al. [27] designed a local transformer with a spatial partition restore network to capture global context dependencies, improving the model's ability to understand complex spatial relationships. Roy et al. [28] introduced the morphological transformer, which enhanced the interaction of shape and structure information, leading to better feature extraction from HSI data. Lightweight models have also been explored to reduce computational complexity while maintaining performance. Scheibenreif et al. [29] built a large dataset to systematically investigate transformer architectures, providing a foundation for lightweight model development. Zhao et al. [30] proposed a multiattention transformer to improve the correlation of center points and their surroundings, enhancing the model's ability to capture fine-grained details. Zhang et al. [31] developed a lightweight transformer network to associate each pixel with global information, making the model more efficient and effective for large-scale HSI data. Wang et al. proposed an end-to-end method that effectively encodes information using transformers, streamlining the classification pipeline and improving performance [32]. Zhang et al. introduced a dimension-driven multipath attention residual block to gain multiscale features, with a channel attention mechanism to improve data depth features [33]. Zhang et al. combined multiattention mechanisms and transformers in their model, which enhanced the model's ability to integrate multiple sources of information for better classification results [34].

Although transformer-based classification methods perform well, their computational complexity is high. Given the constraints of computing resources on mobile terminals, the emphasis on lightweight networks revolves around maintaining algorithm effectiveness while minimizing network complexity. In addressing the challenge of limited sample sets, a lightweight CNN [35] was proposed. Leveraging spectral–spatial Schroedinger eigenmaps, this approach extracts both spectral and spatial features and feeds them into a specialized CNN architecture. To tackle the issue of limited training examples, a multiscale CNN [36] was introduced, aiming to extract multiscale information for enhanced model performance. For HSI classification, an efficient patch-free global learning framework was devised to reduce the computational burden associated with patch-based learning frameworks. Zheng et al. [37] proposed a fast patch-free global learning method to solve the divergence problem and maintain the ability of global spatial information extraction. Wang et al. [38] combined original information with abstract hierarchical features, enabling information compensation to derive low-level and semantic features.

Although current deep learning algorithms have demonstrated reasonable performance in HSI classification, several issues persist.

1) The contribution of different channels to the discriminative information is nonuniform. And usually the most important information is distributed in a few key channels. At the same time, HSIs have rich spectral information. For classification, the contribution of similar bands is similar, while the amount of information contained in different bands varies, which needs to be separated and reaggregated to obtain HSI features with uniform information distribution.
2) Information unrelated to the central pixel in neighboring blocks interferes with classification performance. Hence, heterogeneous pixels may interfere with the center pixel and reduce the accuracy of classification. Closer pixels are more likely to be the same class as the center pixel [22]. Most of the neighboring points in the block correspond to the same objects and provide very meaningful information with some redundancy. Therefore, it is necessary to assign different weights to different pixels and remove redundant pixels to reduce the amount of calculation.

To solve the above problems, we propose an information redistribution module (IRM) and weighted sparse attention module (WSAM) specifically for HSI classification. IRM is composed of an information separation part and an information reaggregation part, which separate the information density degree and reaggregate the feature information of the HSIs, respectively. The IRM alleviates the problem of uneven information distribution inside the feature module and reduces the channel dimension, computational cost, and redundancy in storage. The weighted sparse attention operation prioritizes pixel information close to the center pixel in the HSI, while sparsification is performed to minimize computational cost and storage requirements. In addition, a two-layer convolution

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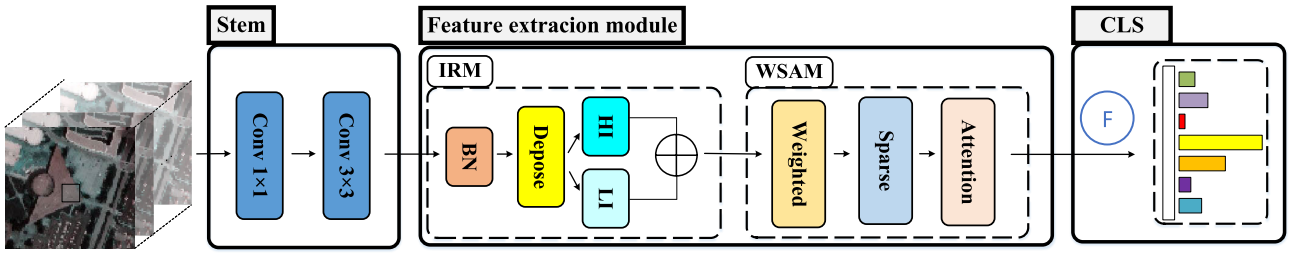


Fig. 1. Illustration for the proposed LIRnet, which includes a simple stem, IRM, WSAM, and a classification head (CLS). F is full connection operation. HI and LI are high information entropy and low information entropy features, respectively.

module was used for feature extraction. The number of feature channels is reduced to 64, which still exceeds the number of sensitive bands, and sensitive bands refer to spectral bands that have a strong ability to distinguish specific categories. In general, the lightweight design of all the modules and low-dimensional feature vectors significantly reduces the computational complexity.

The main contributions of this work are summarized as follows.

1) The IRM is proposed to separate and reaggregate the information of different channels to make the distribution of feature information more uniform and reduce computational overhead.
2) The WSAM is proposed to be more suitable for the lightweight attention operation of hyperspectral data and effectively extract features from HSIs.
3) The overall network module focuses on lightweight design, which can be effectively deployed in edge computing applications.

The remainder of this article is organized as follows. Section II explicitly explains the proposed lightweight HSI classification-based information redistribution network (LIRnet). Section III displays the experimental results to demonstrate the superior performance of LIRnet. In Section IV, we conclude this article.

II. Proposed Classification Framework

The proposed LIRnet is illustrated in Fig. 1 including a simple stem, IRM, WSAM, and a classification head.

A. Simple Stem

In this stem, a 1×1 convolution is used to reduce the dimension and decrease subsequent calculations while extracting spectral features, and a 3×3 convolution is used to extract spectral and spatial features. For the import 1×1 convolution, a 3-D tensor including all pixels of the input patch is mapped into a new 3-D tensor in the channel direction to extract the features among the spectral dimensions. This tensor is used for convolution and attention operations to extract local features and global features.

1) Computational Effort of the Convolution Operation: For standard convolution that has the kernel $K_i \in \mathbb{R}^{C_{\text{in}} \times C_{\text{out}} \times k \times k}$, where C_{in} and C_{out} are the input and output channel sizes, respectively, and k is the kernel size. We choose the k As 11, and k must be odd for patch-based methods. Suppose $\mathbf{F} \in \mathbb{R}^{C_{\text{in}} \times H \times W}$ and $\mathbf{G} \in \mathbb{R}^{C_{\text{out}} \times H \times W}$ are the input and output tensors, respectively, H and W mean the height and width, respectively, and $\mathbf{f}_{\text{ij}} \in \mathbb{R}^{Cin}$ and $\mathbf{g}_{\text{ij}} \in \mathbb{R}^{Cout}$ are the vectors of pixel (i, j) corresponding to $\mathbf{F}$ and $\mathbf{G}$, respectively. The standard convolution is defined as

$$\mathbf{g}_{i,j} = \sum_{p,q} \mathbf{K}_{p,q} \mathbf{f}_{i+p-\frac{k}{2}, j+q-\frac{k}{2}} \tag{1}$$

where $\mathbf{K}_{p,q} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}}}$, $p, q \in 0, 1, \ldots, k-1$ indicates the kernel weights about the kernel position indices (p, q).

The number of parameters of standard convolution is $\text{params} = C_{\text{out}} \times C_{\text{in}} \times k \times k$. The computation of standard convolution is $\text{comps} = H \times W \times C_{\text{out}} \times (k \times k \times C_{\text{in}} + (k \times k - 1) \times C_{\text{in}} + (C_{\text{in}} - 1))$, where $k \times k \times C_{\text{in}}$ means the computation of multiplication, and $(k \times k - 1) \times C_{\text{in}} + (C_{\text{in}} - 1)$ means the computation of addition for each output point. This stem uses a 1×1 convolution that reduces the spectral dimension to 64-D, reducing the number of parameters and the amount of computation by $36\times$ compared with the common operation of 3×3 convolution extending to 256-D. The whole network controls the spectral dimension to 64-D. Compared with the common 256-D and 512-D, the number of parameters and the calculation amount of convolution operation with the same convolution kernel are reduced by 16–$64\times$.

2) Computational Effort of the Transformer Block: The transformer block has two feature extraction modules: the attention module and the feedforward network (FFN). We take a standard self-attention block as an example. The input and output tensors are $\mathbf{F} \in \mathbb{R}^{C_{\text{in}} \times H \times W}$ and $\mathbf{G} \in \mathbb{R}^{C_{\text{out}} \times H \times W}$, respectively, and it has N heads. Then, the first step operation of attention can be represented as

$$\mathbf{q}_{i,j} = \mathbf{ReLU}\left(\mathbf{W}_{o1}\mathbf{x}_{i,j} + b_{o1}\right) \tag{2}$$

$$\mathbf{k}_{i,j} = \mathbf{ReLU}\left(\mathbf{W}_{o2}\mathbf{x}_{i,j} + b_{o2}\right) \tag{3}$$

$$\mathbf{v}_{i,j} = \mathbf{ReLU}\left(\mathbf{W}_{o3}\mathbf{x}_{i,j} + b_{o3}\right) \tag{4}$$

where $\mathbf{x}_{i,j}$ is the input feature of pixel (i, j), $\mathbf{W}_{o1}$, $\mathbf{W}_{o2}$, and $\mathbf{W}_{o3}$ are the projection matrices for queries, RELU is the nonlinear activation function, and $\mathbf{q}_{i,j}$, $\mathbf{k}_{i,j}$, and $\mathbf{v}_{i,j}$ are the intermediate features. The next operation of attention can be represented as

$$\mathbf{Attn} = \mathbf{SoftMax}\left(\frac{\mathbf{q}_{i,j}^T \mathbf{k}}{\sqrt{\text{length}(\mathbf{q}_0)}}\right) \tag{5}$$

where **SoftMax** is the nonlinear activation function, and **Attn** is the attention weight matrix which represents the relationship

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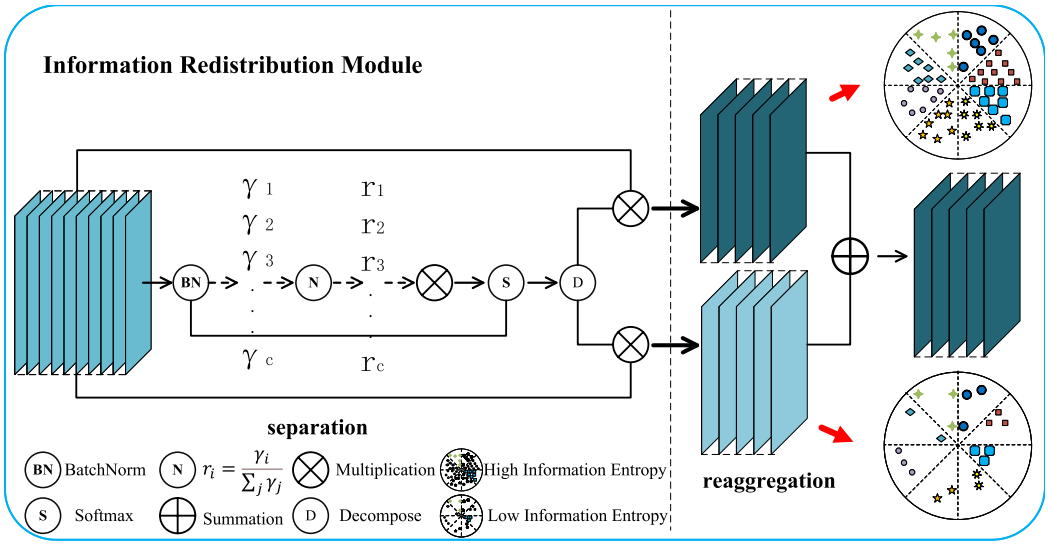


Fig. 2. Illustration for the essential structure of IRM. The high information entropy represents features with high information entropy. And low information entropy is feature with low information entropy.

between the point (i, j) and other points in the feature map. Then, the final operation can be represented as

$$\mathbf{g}_{\mathrm{ij}} = \mathbf{Concat}_{1,\ldots N}\left(\sum_{p,q} \mathbf{Attn}\, \mathbf{v}_{p,q}\right) \tag{6}$$

where points (p, q) belong to the neighborhood of point (i, j), and N is the number of heads. The FFN can be represented as

$$\mathbf{y}_{i,j} = \mathbf{W}_2\, \mathbf{ReLU}\left(\mathbf{W}_1 \mathbf{x}_{i,j} + b_1\right) + b_2 \tag{7}$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ are the projection matrices. The number of parameters in a standard attention operation is params $= 3 \times C_{\mathrm{out}} \times C_{\mathrm{in}}$. The computation of a standard attention operation is comps $= 3 \times C_{\mathrm{in}} \times d + 2 \times n^2 \times d$, where n is the number of points and d is the number of output channels. After the channel dimension is fixed to 64, the number of parameters and computation amount of this operation is reduced by more than four times compared with the commonly used 256 channels. The number of parameters in a standard fully connected operation in the transformer block is params $= C_{\mathrm{out}} \times C_{\mathrm{mid}} \times C_{\mathrm{in}}$. The computation of a standard fully connected operation in the transformer block is comps $= n \times (C_{\mathrm{out}} \times C_{\mathrm{mid}} + C_{\mathrm{mid}} \times C_{\mathrm{in}})$. After the channel dimension is fixed to 64, the number of parameters and computation amount of this operation are reduced by 16× and 4×, respectively, compared with the commonly used 256 channels.

B. Information Redistribution Module

As shown in Fig. 2, IRM is designed to separate and reaggregate information. IRM includes a batch normalization (BN) layer and a feature separation and reaggregation operation. The feature information is filtered by the scale factor in BN layer and is separated into two parts: the high information entropy feature and the low information entropy feature. Because information entropy represents the average amount of information, the upper and lower information circles in the reaggregation part of Fig. 2 represent the amount of information in the respective channels. The high information entropy circle represents features with high information entropy, that is, features that contain more information. The low information entropy circle is the low entropy feature, that is, the feature that contains less information. Then, they are reaggregated to obtain features with uniform information distribution, and the feature channels are reduced by half. The uniform information entropy distribution enables the model to learn each feature in a balanced way, avoiding the model's excessive dependence on certain features or channels. Such balanced learning helps improve the generalization ability of the model, making its performance more consistent between training and testing. At the same time, the number of channels is reduced by half, which in turn reduces the computational complexity.

The features are grouped along the channel direction. The principle involves gradient backpropagation and information entropy theory. The BN layer is the key. The first block is the BN block. In neural networks, gradient flow is a crucial mechanism for parameter updates. For a BN layer, the parameters γ and β are updated during backpropagation based on the gradients of the loss function concerning these parameters.

Assuming we have a BN layer with input x and output y, the forward propagation formulas are

$$\hat{x} = \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} \tag{8}$$

$$y = \gamma \hat{x} + \beta \tag{9}$$

where γ is the scaling factor and β is the shift factor, After normalization, the distribution of the feature map changes to a normal distribution with a mean of 0 and a variance of 1. γ adjusts the scale of the normalized feature values, and β adjusts the mean of the normalized features to accommodate the training of the network. During backpropagation, we need

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to compute the gradient of the loss function L concerning γ

$$\frac{\partial L}{\partial \gamma} = \sum_i \frac{\partial L}{\partial y_i} \cdot \hat{x}_i. \tag{10}$$

From this formula, it can be seen that the gradient of γ is proportional to the normalized input $\hat{x}$, which means that the value of γ reflects the contribution of the input feature to the loss function.

In information theory, the amount of information can be measured by entropy. Suppose the entropy of the input feature after normalization $\hat{x}$ is $H(\hat{x})$, and the entropy of the feature after the BN layer is $H(y)$. The normalization process of the BN layer does not change the relative distribution of the input features but adjusts their mean to 0 and variance to 1.

After normalization, the BN layer applies the scaling factor γ and the shift factor β

$$y = \gamma \hat{x} + \beta. \tag{11}$$

Since β is a shift, it does not affect the entropy of the distribution. The scaling factor γ changes the variance of the distribution. If the variance of $\hat{x}$ is 1, then the variance of y is γ^2.

The entropy of a scaled random variable can be derived from the property of entropy under linear transformations. If $Z = aX$, where a is a constant, then the entropy $H(Z)$ is given by

$$H(Z) = H(aX) = H(X) + \log|a|. \tag{12}$$

Applying this to the above case

$$H(y) = H\left(\gamma \hat{x}\right) = H\left(\hat{x}\right) + \log|\gamma| \tag{13}$$

we get

$$H(y) = H\left(\hat{x}\right) + \log|\gamma|. \tag{14}$$

$\hat{x}$ is a standard normal distribution, and the entropy of a standard normal distribution is a constant. From the above derivation, we can conclude that the size of the scaling factor γ directly affects the entropy of the output feature $H(y)$. In information theory, larger entropy corresponds to a distribution with more information content. A larger γ value increases the entropy of the output feature, indicating that the output feature contains more information. The information of output feature $H(Y)$ is proportional to the logarithm of the scaling factor γ, which means that the larger the γ, the more information of output feature.

Thus, we have strictly proven that in BN, the size of the scaling factor γ indeed represents the amount of information in the corresponding feature: $H(y) = H(\hat{x}) + \log|\gamma|$, which establishes the mathematical relationship between the scaling factor γ and the information content measured by entropy.

First, through the BN layer, the scaling factors are extracted according to the channel direction: $\gamma_1, \gamma_2, \gamma_3, \ldots, \gamma_{64}$, which represent the scaling factors of the ith channel, calculated by the following equation:

$$\gamma_i = \text{Mean}_{p,q}\left(\gamma_{i,p,q}\right) \tag{15}$$

where p and q indicate the position indices (p, q) on each channel. With scaling normalization

$$r_i = \frac{\gamma_i}{\sum_j \gamma_j} \tag{16}$$

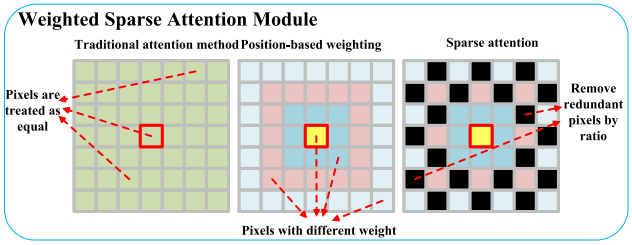


Fig. 3. Illustration for WSAM. The sparse operation removes points of feature blocks before the attention operation.

where $j \in 0, 1, \ldots, 64$

$$s_i = \textbf{SoftMax}(r_i). \tag{17}$$

Then, the final information factors are obtained by **SoftMax**: $s_1, s_2, s_3, \ldots, s_{64}$. Divide $s_1, s_2, s_3, \ldots, s_{64}$ equally into two parts in situ according to numerical magnitude. That is, they are split into two parts according to numerical magnitude while keeping the relative position in each part constant. The corresponding channels are rearranged according to this order. The features are weighted by the corresponding channel through the information factor and then divided into high information entropy features and low information entropy features.

Finally, the features with high information entropy and low information entropy are fused to reduce the maximum gap of information entropy between channels, resulting in features with a more uniform information entropy distribution across channels. The uniform information entropy contributes to stable gradient computation, avoiding the problems of gradient vanishing or gradient explosion. This makes model training more stable and accelerates the convergence process. At the same time, the uniform information entropy ensures that the model does not favor specific channels during the learning process. In this way, each channel can fully contribute, and the model can learn from diverse and balanced information, understand the characteristics of the input data more comprehensively, and improve classification performance.

C. Weighted Sparse Attention Module

In patch-based HSI classification tasks, the information usually provided by certain key spectral bands or the spectra of key pixels can lead to good classification performance. This suggests that there is some redundancy in the spectral information in a single block of the HSI classification task. Moreover, the computational complexity of computing the attention matrix for all points within a block is $O(n^2)$. Thus, using sparse HSI features can significantly reduce the computational burden while maintaining accuracy.

In addition, as shown in Fig. 3, the spectral information carried by each pixel in the patch has different effects on the central pixel. Generally, objects with similar geographical location distribution are more likely to provide information about the type of similar objects. Similarly, closer pixels should provide more information about the central pixel. Conversely, more distant pixels provide less information. Therefore, in the process of calculating the attention matrix, weighting according to different geographical locations can help the attention matrix better learn the relationships between

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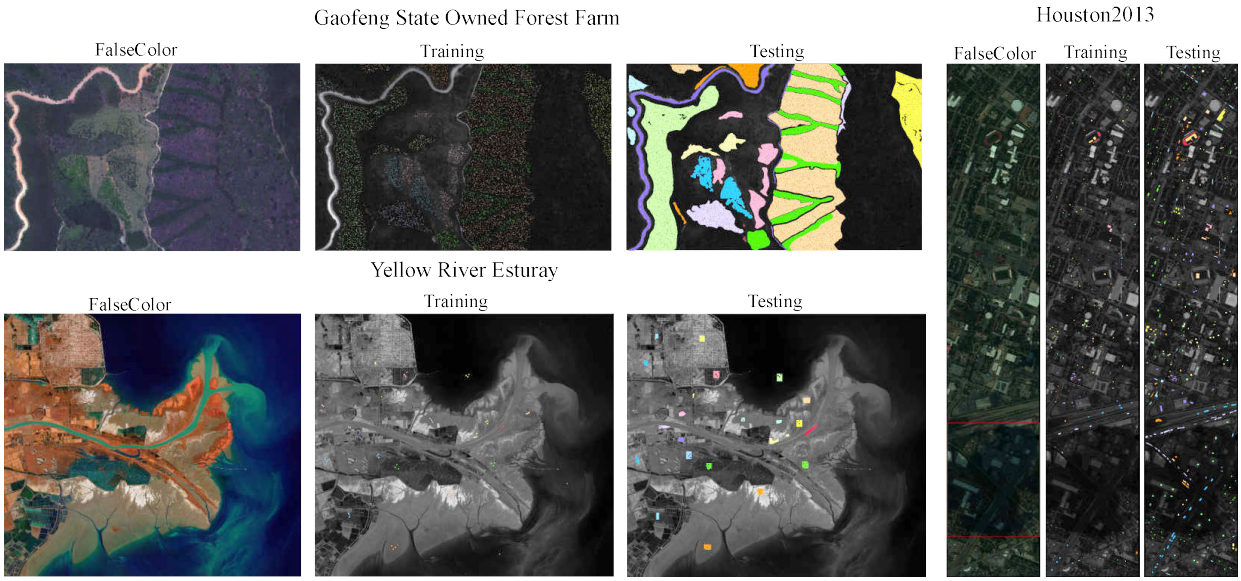


Fig. 4. Maps of false color and training and testing samples' distributions of the GSOFF, H2013, and YRE datasets, respectively.

different pixels, thereby enhancing the performance of classification tasks.

Therefore, we propose the WSAM. The input and output channels of the 3-D tensor are 32 in the transformer block.

For the input features, each feature point is weighted according to its distance to the central feature, making the learning of the attention matrix more efficient. Feature points that are closer to the central feature point receive larger weights, while feature points that are farther away receive lower weights. At the same time, within points of different distances, the same proportion of points are randomly removed to reduce data redundancy and the amount of computation. This is done as follows:

$$\mathbf{x}_{i,j} = \left(1 - \boldsymbol{\mu} \times \mathbf{Dis}\left(\mathbf{x}_{i,j}, \mathbf{x}_o\right)\right) \times \mathbf{x}_{i,j} \tag{18}$$

$$\mathbf{Dis}\left(\mathbf{x}_{i,j}, \mathbf{x}_o\right) = \max(|i - \mathbf{o}_0|, |j - \mathbf{o}_1|) \tag{19}$$

where $\boldsymbol{\mu}$ is the hyperparameter that controls the weight of feature points, $\mathbf{x}o$ is the central feature point, $\mathbf{Dis}(\cdot)$ is the distance between $\mathbf{x}_{i,j}$ and $\mathbf{x}_o$, and $\mathbf{o}_0$ and $\mathbf{o}_1$ are the position indices of $\mathbf{x}_o$

$$\mathbf{x}_{i,j} = \begin{cases} \mathbf{x}_{i,j}, & \text{if random}(\cdot) < \boldsymbol{\epsilon} \\ \mathbf{remove}, & \text{else} \end{cases} \tag{20}$$

where $\boldsymbol{\epsilon}$ is the hyperparameter that determines the number of feature points to remove, and random$(\cdot)$ is a random function that produces a random real number between 0 and 1. The linear projection in attention also fixes the number of channels. In the fully connected layer of the transformer, the middle dimension is raised to 128 and then lowered back to 32. The height and width are h and w, respectively, and the number of channels is 32. The h and w are odd numbers, and $h = w$. Through (4)–(6), the attention features are obtained.

The prediction probability vector in classification is defined as

$$\mathbf{y}_{i,j} = \mathbf{W}_2\, \mathbf{ReLU}\left(\mathbf{W}_1 \mathbf{x}_{i,j} + b_1\right) + b_2, \tag{21}$$

where $\mathbf{W}_1 \in \mathbb{R}^{32\times128}$ and $\mathbf{W}_2 \in \mathbb{R}^{128\times32}$ are the projection matrices, and $b_1 \in \mathbb{R}^{128}$ and $b_2 \in \mathbb{R}^{32}$ are the bias vector of projection matrices $\mathbf{W}_1$ and $\mathbf{W}_2$. Then, we transfer the 2-D matrix back to a 3-D tensor and transfer the 3-D tensor to a 1-D vector by adaptive pooling in PyTorch. The final result is obtained using the 1-D vector through a fully connected layer.

The cross-entropy function used in the final classification task is defined as

$$\mathcal{L}_{cls} = -\sum_{c=1}^{C} e_i^c \log p_i^c \tag{22}$$

where e_i^c is the ground truth, and p_i^c is the final prediction probability of the ith category.

III. Experiments and Analysis

In this section, the effectiveness of the proposed LIRnet is evaluated using three HSI datasets. Initially, the section presents three HSI datasets: Gaofeng state owned forest farm (GSOFF), Houston2013 (H2013), and yellow river estuary (YRE), each representing diverse scenarios including forest, urban, and moist soil, respectively. Subsequently, ablation tests are performed to validate the efficacy of the suggested LIRnet. Following this, experiments involving LIRnet are compared against various conventional and state-of-the-art algorithms. Finally, the operational efficiency of LIRnet and other algorithms is assessed on an embedded device to confirm their performance.

A. Datasets

1) Gaofeng State Owned Forest Farm: In January 2018, the AISA Eagle II diffraction grating push-broom hyperspectral imager system captured a dataset over a farm in Guangxi, China. The dataset comprises 572 × 906 pixels, with a spatial resolution of 1 m and a spectral resolution of 3.3 nm. It encompasses 125 spectral bands within the wavelength range of 0.4–1.0 μm. Specifically, it delineates nine distinct forest vegetation classes along with three other classes. Table I illustrates the distribution of training and testing samples, while Fig. 4 displays the false-color map and sample distribution.

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TABLE I
CLASS LABELS AND TRAIN–TEST SAMPLES' DISTRIBUTION OF GSOFF, H2013, AND YRE

No.	GSOFF			H2013			YRE		
	Class Name	Train	Test	Class Name	Train	Test	Class Name	Train	Test
1	Cuninghamia lanceolata	4038	76705	Grass healthy	198	1053	Spartina alterniflora	84	743
2	Pinus massoniana	268	5072	Grass stressed	190	1064	Low-tide mudflat	84	709
3	Pius elliottii	208	3967	Grass synthetics	192	505	High-tide mudflat	91	645
4	Eucalyptus grandis x urophylla	746	14185	Tree	188	1056	Mixed area 1	16	123
5	Eucalyptus urophylla	2014	38261	Soil	186	1056	Mixed area 2	95	774
6	Catanopsis hystrix	1274	24187	Water	182	143	Mixed area 3	75	731
7	Mytilaria laosensis	150	2850	Residential	196	1072	Oil field	72	612
8	Camellia oleifera	416	7908	Commercial	191	1053	Sea	135	756
9	Other broadleaf forest	538	10203	Road	193	1059	Intertidal phragmite	86	758
10	Road	524	9948	Highway	191	1036	Ecological reservoir	75	563
11	Cutting blank	508	9662	Railway	181	1054	Suaeda salsa	67	610
12	Building land	18	50	Parking lot 1	192	1041	Salt fields	81	663
13				Parking lot 2	184	285	Arable land	68	681
14				Tennis court	181	247	Phragmite	83	756
15				Running track	187	473	Woodland	49	502
16							Lotus pond	75	461
17							Typha orientals presl	6	30
18							Robina pseudoacacia	65	516
19							Pond	86	763
20							Yellow river	88	771
	Total	10702	202998	Total	2832	12197	Total	1481	12167

TABLE II
ANALYSIS EXPERIMENTS ABOUT THE PARAMETER μ OF WSAM-W IN REFERENCE TO THE CLASSIFICATION PERFORMANCE (OVERALL ACCURACY [%]) FOR THE GSOFF, H2013, AND YRE DATASETS

μ	0.2	0.15	0.1	0.05	0.01
GOSEF	95.55	94.37	**97.56**	96.97	95.10
H2013	84.16	83.99	**89.56**	88.71	88.80
YRE	90.65	91.47	**93.74**	92.19	93.00

TABLE III
ANALYSIS EXPERIMENTS ABOUT THE PARAMETER ϵ OF WSAM-S IN REFERENCE TO THE CLASSIFICATION PERFORMANCE (OVERALL ACCURACY [%]) FOR THE GSOFF, H2013, AND YRE DATASETS

ϵ	0.4	0.5	0.6	0.7	0.8
GOSEF	97.13	**97.56**	95.33	96.01	92.55
H2013	87.36	**89.56**	84.81	82.63	83.01
YRE	93.41	**93.74**	92.00	91.83	90.27

2) Houston2013: In June 2012, the CASI-1500 sensor captured a dataset comprising 349 × 1905 pixels over the University of Houston and its surrounding areas. The dataset features a spatial resolution of 2.5 m and a spectral resolution of 4.65 nm, encompassing 144 spectral bands within the wavelength range of 0.38–1.05 μm. This dataset encompasses ten man-made and five natural objects. Table I outlines the distribution of training and testing samples for each class, accompanied by the false-color map and sample distribution. Notably, the red region in the false-color map is affected by cloud cover.

3) Yellow River Estuary: In November 2018, the GF-5 satellite captured a dataset over the yellow river delta, comprising 1185 × 1342 pixels with a spatial resolution of 30 m. This dataset offers a spectral resolution of 5 nm, encompassing 150 spectral bands within the VNIR range of 0.40–1.00 μm and a spectral resolution of 10 nm containing 180 spectral bands within the SWIR range of 1.00–2.50 μm. Notably, bands 1 and 2 in VNIR, as well as bands 175–180, 172–173, 119–121, 96–115, and 42–53 in SWIR, were excluded, resulting in a total of 285 bands. The dataset encompasses 20 wetland classes, and Table I presents the distribution of training and testing samples. In addition, visual representations include false-color maps and a display of the sample distribution.

TABLE IV
ABLATION EXPERIMENT

IRM	μ	ϵ	H2013	GSOFF	YRE
×	×	×	75.40	92.17	85.88
✓	×	×	83.19	95.48	90.30
×	✓	×	84.84	95.13	90.17
×	×	✓	80.36	95.31	88.61
✓	✓	×	86.99	95.91	91.87
✓	×	✓	87.47	96.89	92.39
×	✓	✓	88.51	96.95	92.62
✓	✓	✓	**89.56**	**97.56**	**93.74**

B. Parameter Analysis

The method applied to the adjustable parameters within LIRnet involves the sparse parameter ϵ and the weighting parameter μ in WSAM. As shown in Tables II and III, different parameters produce different effects, all showing changes in classification performance. The two parameters affect the performance of the network from different aspects, so the optimal data range is not on the same scale, and further refinement is needed to optimize the results. Therefore, although the adopted IRM and WSAM methods effectively

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TABLE V
CLASSIFICATION RESULTS [%] OF GSOFF WITH DIFFERENT METHODS

Class No.	OTVCA [39]	CNNHSI [40]	RPCACNN [7]	SSCNN [8]	CDCNN [9]	DRCNN [10]	BERTHSI [23]	FusCNNGCN [41]	3DCNN [13]	LIRnet
1	95.34	86.83	83.48	91.61	92.05	**98.00**	96.99	95.78	95.43	97.55
2	**99.70**	11.73	34.27	75.51	77.43	92.61	92.94	89.33	92.84	97.61
3	99.24	3.05	20.04	85.53	73.56	92.74	93.80	84.93	97.08	**99.95**
4	**100**	77.99	52.23	86.07	91.62	99.28	97.46	98.06	96.84	**100**
5	99.93	61.86	66.16	96.73	99.09	99.58	92.44	98.48	97.51	**99.99**
6	83.31	45.59	29.80	85.07	76.52	86.41	90.09	83.82	84.41	**92.01**
7	**99.89**	1.16	6.91	70.28	73.86	96.91	83.58	92.67	94.28	97.82
8	99.03	58.80	72.03	95.52	94.92	99.06	**99.97**	97.77	98.47	99.42
9	91.30	45.53	54.82	68.81	77.59	89.16	93.46	83.40	90.74	**93.58**
10	98.43	97.63	98.13	99.10	99.66	**99.76**	99.69	99.49	99.01	97.84
11	**99.98**	94.78	95.74	99.15	99.22	99.83	98.34	99.81	99.69	99.73
12	0.00	7.81	0.00	90.63	65.63	26.56	68.75	48.44	45.31	**100**
OA	95.63	69.59	67.48	90.32	90.60	96.50	95.12	94.42	94.80	**97.56**
AA	88.85	49.40	51.13	87.00	85.09	89.99	92.29	89.33	90.97	**97.96**
Kappa*100	94.48	60.91	58.69	87.79	88.11	95.57	93.84	92.95	93.45	**96.93**

TABLE VI
CLASSIFICATION RESULTS [%] OF H2013 WITH DIFFERENT METHODS

Class No.	OTVCA [39]	CNNHSI [40]	RPCACNN [7]	SSCNN [8]	CDCNN [9]	DRCNN [10]	BERTHSI [23]	FusCNNGCN [41]	3DCNN [13]	LIRnet
1	80.72	82.53	82.34	82.53	82.43	82.34	**83.00**	82.62	80.63	82.62
2	79.04	83.55	82.33	**84.96**	**84.96**	84.77	83.36	78.10	**84.96**	84.02
3	99.21	99.60	85.35	98.81	**100**	88.71	91.09	91.68	95.84	99.41
4	84.75	**98.58**	91.10	92.23	93.37	92.33	93.56	96.31	71.40	88.16
5	98.20	98.67	93.47	99.24	100	99.91	99.81	**100**	90.06	97.92
6	94.41	99.30	83.92	93.01	95.80	99.30	98.60	98.60	**100**	95.80
7	89.93	71.08	74.16	94.12	79.38	89.09	87.22	91.23	**93.56**	74.91
8	49.76	62.58	63.34	68.95	71.23	67.90	68.76	85.85	85.57	**89.93**
9	85.08	59.87	76.02	80.64	78.75	78.00	77.15	72.71	72.71	**83.95**
10	67.37	96.24	46.04	43.24	49.71	63.51	48.46	61.49	42.57	**99.61**
11	74.10	81.69	66.03	59.49	79.22	72.68	82.73	79.51	65.65	**86.62**
12	80.02	61.19	60.42	90.11	78.39	91.64	86.17	92.03	78.48	**98.56**
13	58.25	59.30	78.95	85.61	**92.63**	89.47	86.32	90.53	67.02	78.95
14	100	100	98.38	95.95	100	100	99.60	99.60	99.19	**100**
15	97.25	98.10	93.23	94.50	100	83.09	98.94	90.70	84.36	**100**
OA	80.62	81.30	75.60	81.63	82.32	83.28	82.90	85.23	78.25	**89.56**
AA	82.54	83.49	78.34	84.23	85.73	85.52	85.65	87.40	80.80	**90.70**
Kappa*100	79.13	79.69	73.66	80.05	80.95	81.94	81.51	83.96	76.49	**88.65**

improve the classification performance of the network, meticulous parameter tuning is required to obtain optimal results.

C. Ablation Experiment

The objective of the ablation experiments is to verify the effectiveness of the adopted IRM and WSAM. The classification performance was evaluated using overall accuracy (OA). As shown in Table IV, the influence of sparse and weighted operations in IRM and WSAM on the classification performance in HSI classification is included. In LIRnet, the OA of the H2013, GSOFF, and YRE datasets showed significant improvements, increasing by 14.16%, 5.49%, and 7.86%, respectively. Therefore, the effectiveness of the proposed LIRnet in classification is confirmed.

To discuss the effect of IRM on ablation performance, ablation experiments were performed. As shown in Table IV, OA improves by 7.79%, 3.31%, and 4.42% for the GSOFF, H2013, and YRE datasets, respectively, which again verifies the effectiveness of IRM.

To discuss the effect of weight parameter $\boldsymbol{\mu}$ of WSAM (WSAM-W) and sparse rate $\boldsymbol{\epsilon}$ of WSAM (WSAM-S) on ablation performance, ablation experiments were performed. As shown in Table IV, OA improves by 9.44%, 2.96%, 4.29%, and 4.94%, 3.14%, and 2.33% for the GSOFF, H2013, and

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TABLE VII
CLASSIFICATION RESULTS [%] OF YRE WITH DIFFERENT METHODS

Class No.	OTVCA [39]	CNNHSI [40]	RPCACNN [7]	SSCNN [8]	CDCNN [9]	DRCNN [10]	BERTHSI [23]	FusCNNGCN [41]	3DCNN [13]	LIRnet
1	100	100	100	100	100	100	100	100	89.91	**100**
2	100	100	100	100	100	100	100	100	100	**100**
3	100	97.52	100	95.97	99.22	100	100	100	100	**100**
4	**100**	82.93	**100**	**100**	**100**	**100**	**100**	**100**	**100**	96.65
5	44.32	75.19	66.93	**85.66**	20.41	48.84	43.15	25.58	23.00	77.31
6	100	73.60	64.71	81.81	86.05	90.70	100	84.27	74.83	**100**
7	**100**	54.58	46.90	69.61	**100**	97.71	**100**	98.37	**100**	91.83
8	100	100	100	100	100	100	100	100	100	**100**
9	100	99.21	100	100	100	100	100	100	100	**100**
10	100	86.50	78.69	89.52	100	100	100	97.87	100	**100**
11	92.79	96.39	73.11	51.80	91.80	**97.38**	45.57	42.79	89.02	85.28
12	90.95	63.95	65.91	84.16	70.44	73.30	70.29	62.90	**97.74**	95.63
13	48.90	60.35	47.28	46.99	**98.97**	91.04	77.97	99.41	89.57	99.85
14	100	96.69	91.80	89.81	100	100	100	100	100	**100**
15	**100**	79.08	79.88	73.90	86.25	99.60	92.03	82.67	98.80	63.39
16	**95.44**	60.95	30.37	77.22	87.64	86.55	64.64	78.09	84.60	86.55
17	**100**	93.33	10.00	**100**	**100**	**100**	**100**	**100**	**100**	93.33
18	**99.81**	81.98	30.43	33.91	0.00	22.09	70.74	62.79	92.05	83.53
19	100	81.65	85.45	97.64	97.90	100	100	100	100	**100**
20	100	100	100	100	100	100	100	100	100	**100**
OA	92.56	84.73	77.90	84.00	86.57	90.15	87.89	86.09	90.75	**93.74**
AA	93.61	84.20	73.57	83.90	86.93	90.36	88.22	86.74	91.98	**93.67**
Kappa*100	92.13	83.83	76.59	83.06	85.78	89.58	87.19	85.27	90.21	**92.95**

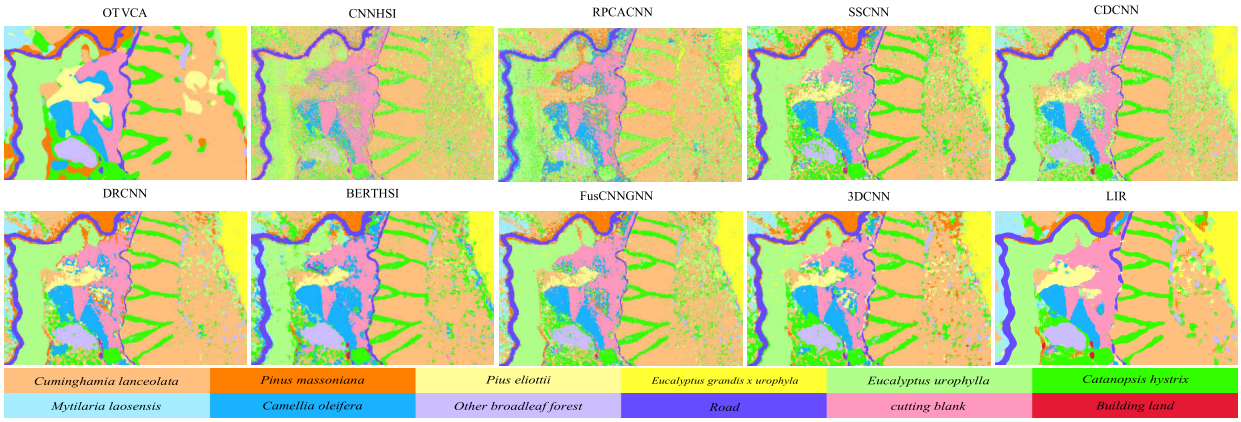


Fig. 5. GSOFF's ground truth and classification maps in different methods.

YRE datasets, respectively, which again verifies the effectiveness of WSAM-W and WSAM-S.

Compared with the results given in Table IV, ablation experiments were performed for all cases to compare the influence of each block on the classification performance. This observation indicates that IRM and WSAM are effective in terms of feature extraction ability. The most significant performance enhancement can be observed in the H2013 dataset. Therefore, IRM and WSAM help the network to better retain discriminative information during the feature extraction process.

D. Classification Performance

In Tables V–VII, class-specific accuracy (CA), OA, average accuracy (AA), and Kappa coefficient are used to evaluate the classification performance. The proposed LIRnet outperforms all the above state-of-the-art techniques by about 1.06% on GSOFF, 4.33% on H2013, and 1.18% on YRE, respectively, outperforming the second-best performing method in OA.

In particular, on the H2013 dataset, LIRnet has the most significant advantage. The reason may be the difference between the test set and the training set of the Houston data, and the objects occluded by clouds present in the test set do not

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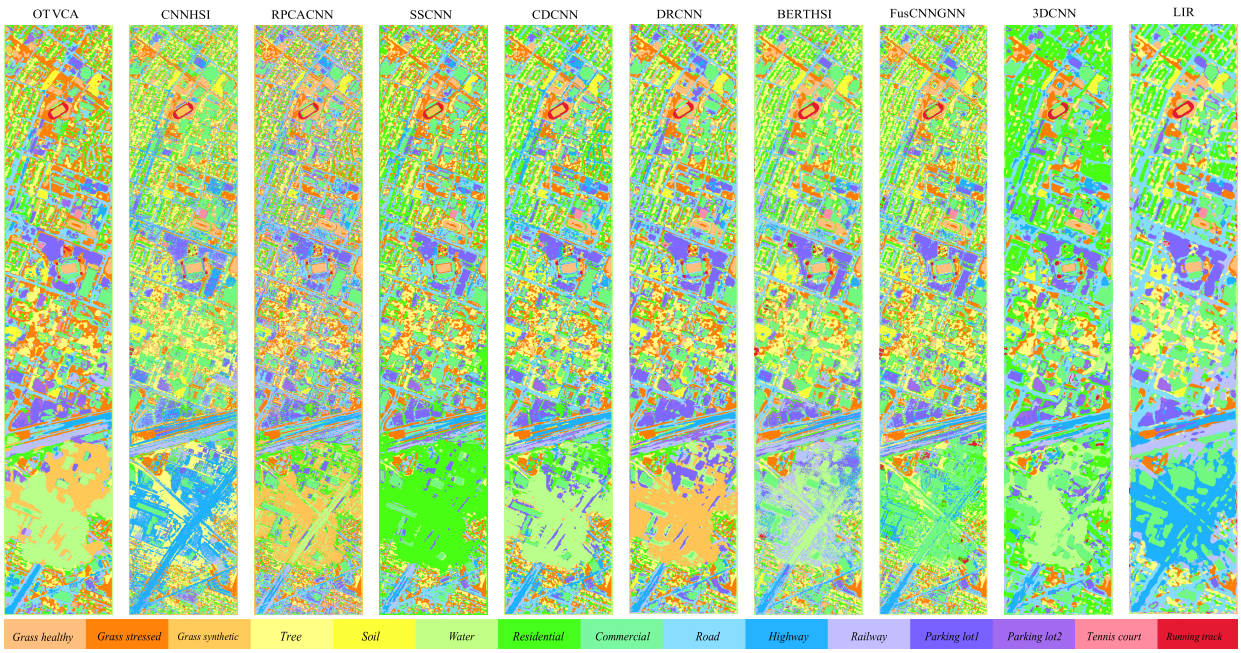


Fig. 6. H2013's ground truth and classification maps in different methods.

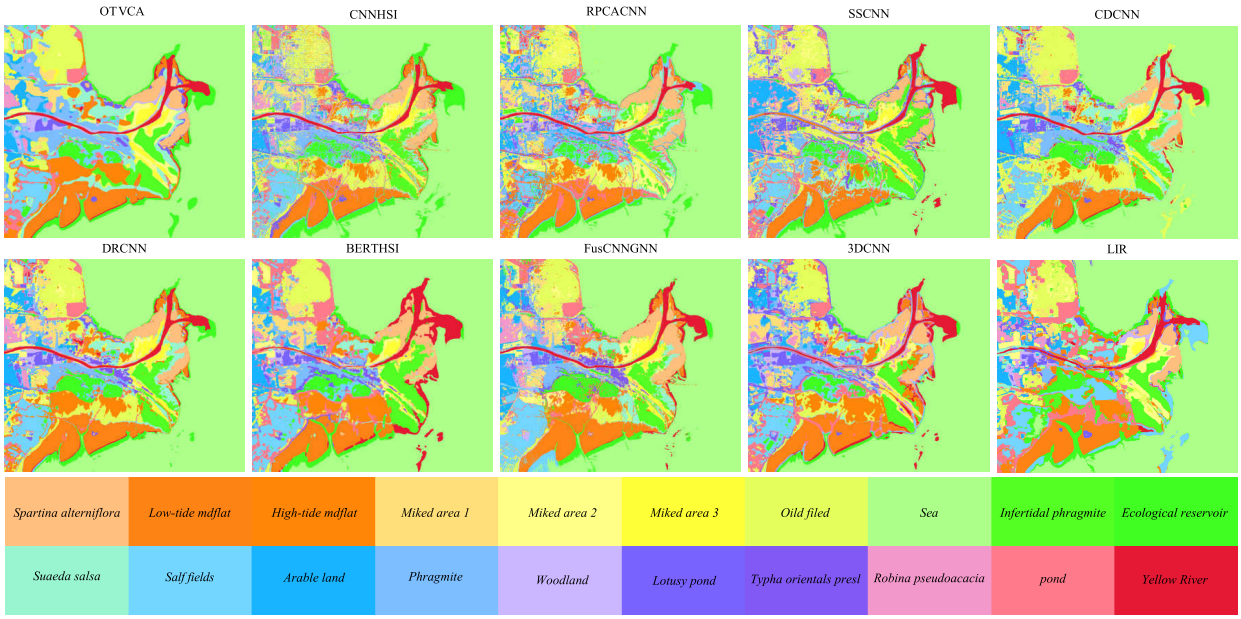


Fig. 7. YRE's ground truth and classification maps in different methods.

appear in the training set. Information homogenization yields a significant improvement on this. This situation indicates that the information of the patch-based hyperspectral classification task itself is enough to train a better network model, and improving the reasonable distribution of information may be more effective than adding more information extraction methods. When training lightweight classification tasks, the lightweight design of the network should be ensured, and the extracted feature information should be rationally used to maximize the classification ability of the network. Thus, as a direct feature extraction method, LIRnet shows superior classification results.

To intuitively illustrate the effect of LIRnet, Figs. 5–7 are used to describe the classification comparison of all the above methods. The results of CNNHSI have obvious salt and pepper noise, and because CNNHSI excludes spatial information, its classification results are the worst, as detailed in Tables V–VII. In contrast, although OTVCA has good classification performance, it is affected by the shadowing of secondary targets. In contrast, LIRnet produces more accurate classification maps compared with the other techniques, exhibiting spatial smoothing and fewer mislabeled pixels, consistent with the tabular results in Tables V–VII. The main challenge of the H2013 dataset is the spectral difference between pixels

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TABLE VIII
INFERENCE TIME OF DIFFERENT METHODS BASED ON NEURAL NETWORK FOR THE GSOFF, H2013, AND YRE DATASETS. UNIT OF TIME IS SECONDS

Method	GSOFF	H2013	YRE
CNNHSI	188	252	897
SSCNN	194	262	938
BERTHSI	310	452	1326
DRCNN	294	384	1412
fusCNNGCN	337	493	1622
CDCNN	409	539	1703
3DCNN	441	630	2730
LIRnet	**98**	**120**	**409**

in the shaded area under the cloud and in different areas, although they belong to the same class. In addition, most of the training data originate from cloud-free regions. Therefore, most advanced methods, except CNNHSI, cannot accurately delineate the highway in the cloud. However, LIRnet can expertly classify these regions. This success is attributed to the reasonably uniform distribution of information, mitigating distribution differences caused by nonuniform data distribution. By capturing the basic attributes of the pixels in the cloud and fog area and homogenizing the distribution of the information in different channels, the method reduces the information difference between the occlusion and the nonocclusion and distinguishes the pixels in the cloud and fog area into a certain extent. In addition, on the remaining two datasets, the classification results show enhanced smoothness and accuracy. But the classification maps of LIRnet are oversmoothing in some boundary, it may be because the IRM module hides some details in the process of improving information utilization. Follow-up work may further improve classification performance by adding modules that strengthen the detail system.

E. Computational Analysis on Embedded Device

LIRnet is designed to ensure that the accuracy of HSI classification tasks can be maintained when deployed in a variety of scenarios with limited computing power. To verify the computational complexity of the proposed LIRnet, each algorithm is implemented using the PyTorch deep learning framework on an NVIDIA DRIVE AGX Orin embedded device. NVIDIA DRIVE AGX features a GPU with 167 TOPS (INT8) and 5.2 TFLOPS (FP32) computational power, along with approximately 200 GB/s (256-bit LPDDR5) memory bandwidth. Table VIII shows the inference times of each algorithm on the above datasets. Among the methods tested on all three datasets, RPCA is too simple a network to extract enough features to complete an effective classification task and is therefore not included in the comparison. The results show that the LIRnet method has a clear advantage in inference speed.

IV. CONCLUSION

This article proposes a lightweight network, LIRnet, for enhancing the rational distribution of feature information in HSI classification. Using IRM and WSAM, this approach aims to strengthen classification performance with a lightweight network. IRM is used to separate and reaggregate the input feature information to obtain features with a uniform information distribution for classification. In addition, WSAM is used to enhance the learning of the attention matrix and greatly reduce the amount of computation. It is worth noting that the proposed method achieves significant classification performance with minimal inference time. Experimental evaluation on three different scenarios of HSI datasets shows the superior performance and better accuracy of the proposed LIRnet compared with the existing HSI feature extraction methods, confirming its effectiveness.

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